Even-Index Representations and Recursive Decomposition of Rectilinear Polygons

Representation, canonicalization, and three-step rectangle decomposition

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Section 1

Outline

Outline

1. Outline
2. Introduction
3. Representation
4. Monotone Hulls
5. Recursive Decomposition
6. Verification
7. Conclusion

Section 2

Introduction

Rectilinear Geometry in Physical Design

- VLSI layouts are overwhelmingly **Manhattan**: every edge is horizontal or vertical.
- Rectilinear polygons model **obstacles**, **regions**, and **layout fracturing** patterns [1], [2].
- Coordinates are **integers**; rounding is not merely inaccurate but illegal.
- The polygons of interest are small: $n \leq 100$ vertices.
- Consequently, **quadratic algorithms are acceptable**; clarity and exactness dominate.

Problem Statement and Contributions

Input. A simple rectilinear polygon P and its even-index encoding $S = (s_0, \dots, s_{m-1})$ with $s_i \in \mathbb{Z}^2$.

Tasks.

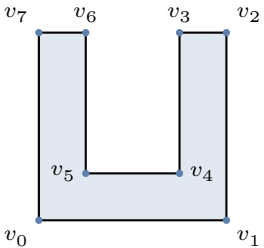
1. A compact even-index representation and its canonical form.
2. An $O(m)$ canonicalization of creation orderings while preserving the perimeter and the area.
3. A three-step rectangle decomposition: convex decomposition, explicit 270° handling, and implicit 270° handling.
4. C++20 implementation and verification.

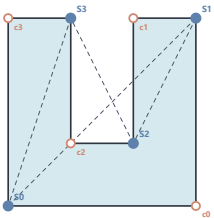
Contributions. The canonical form of the encoding, the compressed area formula, an $O(m)$ canonicalization, a three-step rectangle decomposition, and output-class tests that area checks alone cannot replace.

Section 3

Representation

Full Boundary and Even-Index Encoding

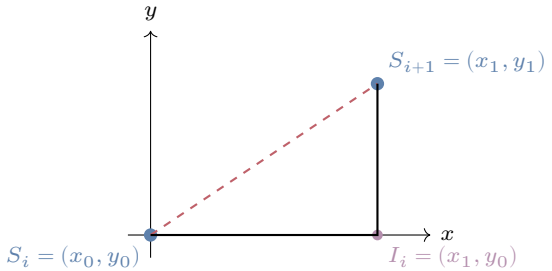




Definition. For a full boundary $V = (v_0, \dots, v_{n-1})$ with n even, the even-index encoding is $S = (v_0, v_2, \dots, v_{n-2})$.

- The odd-index vertices are implicit; only $m = n/2$ points are stored.
- The represented polygon is recovered by the x-first rule below.

x-First Expansion



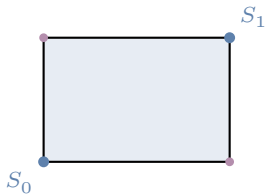
Definition (x-first expansion). For consecutive stored points $S_i = (x_0, y_0)$ and $S_{i+1} = (x_1, y_1)$, insert

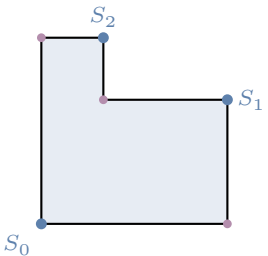
$$I_i = (x_{i+1}, y_i)$$

before S_{i+1}, including the closing pair $S_{m-1} \to S_0$.

- The diagonal chord $S_i S_{i+1}$ is **not** a boundary edge.
- The y-first form (x_0, y_1) reconstructs a different polygon.

Minimal Encodings





- **2 stored points** represent a rectangle: $E(S)$ has four corners.
- **3 stored points** represent a monotone / L-shaped piece: six corners.
- These are not degenerate cases; they are the base cases of the decomposition.

Canonical form. For a simple input polygon, every cyclic consecutive pair of stored points differs in both coordinates; every simple-polygon encoding is canonical. Canonicality is local and does not imply simplicity.

Canonical Form

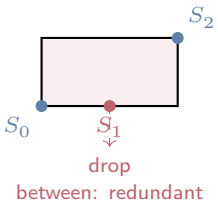
Definition. A stored sequence $S = (s_0, \dots, s_{m-1})$ is **canonical** if

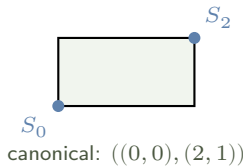
1. $m \geq 2$;
2. no cyclic consecutive points coincide;
3. if $m = 2$, the two points are distinct (a shared axis is allowed);
4. if $m \geq 3$, no point lies strictly between its neighbours on a shared axis.

```
if |S| < 2: return false
if any cyclic consecutive duplicate: return false
if |S| == 2: return true
for each i (cyclic):
    p, q = prev(i), next(i)
    if p.y == s_i.y and x_i strictly between x_p, x_q: return false
    if s_i.x == q.x and y_i strictly between y_p, y_q: return false
return true
```

- Consecutive duplicates are rejected first.
- Non-adjacent duplicates are allowed: canonical, but not simple.

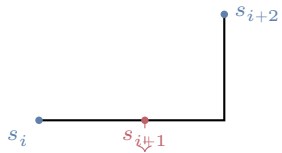
Betweenness Is Redundant



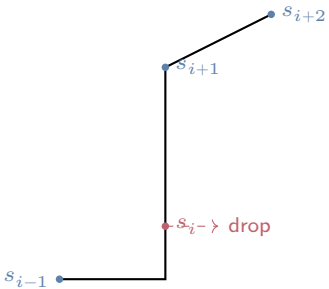


- $S = ((0,0),(1,0),(2,1))$ has S_1 strictly between S_0 and S_2 on the bottom edge.
- Deleting S_1 preserves the boundary, perimeter, and area: the represented polygon is unchanged.
- The two lists have the **same area and perimeter**, so area-only (or perimeter-only) tests cannot detect the redundancy; canonicalization removes it.
- A *spike* is the complementary case: a shared-axis point that is not between its neighbours. Deleting it would shorten the boundary, so it is not redundant.

Canonicalization



same y: drop the second



same x: drop the first

- Remove cyclic consecutive duplicates first.
- A same-y pair drops the second point when it lies strictly between its neighbours; a same-x pair drops the first analogously.
- A work queue revisits only the neighbours of a removed point: $O(m)$.
- The final `is_canonical` check is a safety net.

Orientation and Signed Area

For a simple polygon with vertices (x_i, y_i),

$$A = \frac{1}{2} \sum_i \left(x_i y_{i+1} - x_{i+1} y_i \right).$$

- $A > 0$: anticlockwise; $A < 0$: clockwise.
- The even-index encoding admits an $O(m)$ evaluation over the stored points.
- Perimeter $P(S) = \sum_i (|x_{i+1} - x_i| + |y_{i+1} - y_i|)$; redundant removal preserves both P and A.
- The sign drives the cut branches and serves as a cheap postcondition.

Lemma. The signed area is invariant under the x-first expansion E.

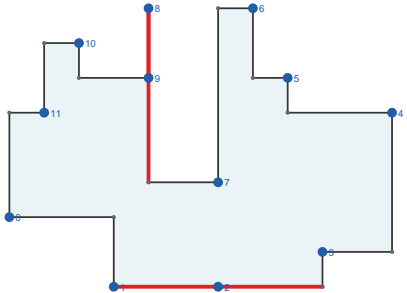
Proof sketch. Each inserted vertex is collinear with an axis-aligned edge and contributes zero net area. □

Creation Outputs Are Not Canonical

rpolygon_xmono: hand-written 12-point fixture

is_canonical = False | red = consecutive same-orientation edges

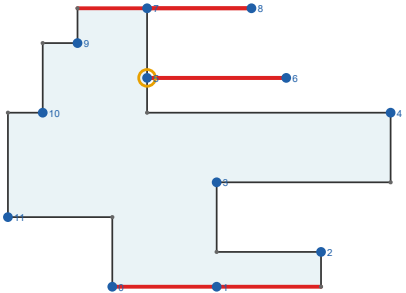
violation 1: (-3, -4) -> (0, -4) -> (3, -4)
violation 2: (-2, -1) -> (-2, 4) -> (-2, 2)



rpolygon_ymono: hand-written 12-point fixture

is_canonical = False | red = consecutive same-orientation edges

violation 1: (-3, -4) -> (0, -4) -> (3, -4)
violation 2: (-2, 2) -> (2, 2) -> (-2, 2)
violation 3: (-2, 4) -> (1, 4) -> (-4, 4)

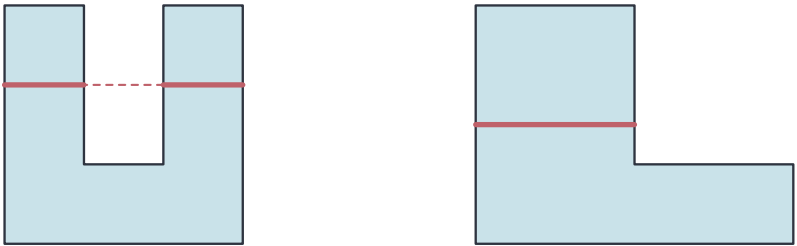


- Raw creation orderings contain consecutive collinear triples.
- Canonicalization removes the redundant points.

Section 4

Monotone Hulls

Monotonicity and Convexity



Definition. A rectilinear polygon is x-monotone if every vertical line meets it in a single interval; y-monotone is defined symmetrically.

Lemma. A rectilinear polygon is orthogonally convex if and only if it is both x-monotone and y-monotone.

Monotone Hull Algorithm

- Let p, c, n be the previous, current, and next stored vertices.
- A vertex is removed when monotonicity is violated and the turn has the wrong sign:

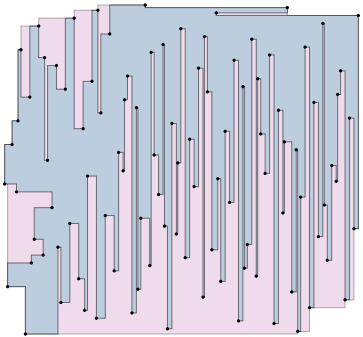
$$\Delta = (y_c - y_p)(x_n - x_c).$$

- A circular doubly-linked list makes removal $O(1)$.
- Two passes: minimum to maximum, then maximum to minimum.

```
for c in chain:
    if dir(c) > dir(n) or dir(p) > dir(c):
        if wrong_sign(Delta):
            remove(c); c = prev(c)
        else:
            c = next(c)
    else:
        c = next(c)
```

Complexity. Each sweep is $O(m)$; the hull's internal restart scan is $O(m^2)$ worst case.

Hull Composition



- x-hull then y-hull yields the rectilinear convex hull.
- Hulls also give monotone enclosures and preprocessing for decomposition.
- The output is checked for canonicality and monotonicity.

Section 5

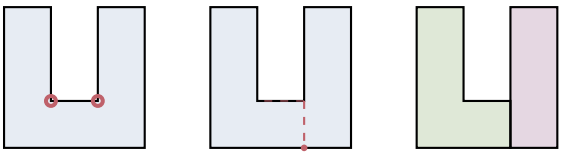
Recursive Decomposition

Three-Step Decomposition



- **Step 1: convex cut.** Remove concavities; simplify the remaining geometry.
- **Step 2: explicit 270°.** Resolve reflex corners that are stored vertices.
- **Step 3: implicit 270°.** Resolve reflex corners introduced by expansion.
- Steps 2 and 3 repeat until neither type remains; the final pieces are rectangles, suitable for layout fracturing and DRC [3], [4].

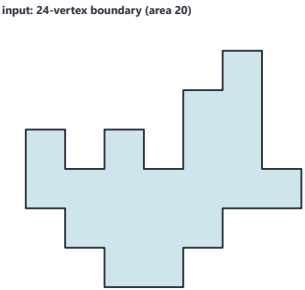
Cut Primitive

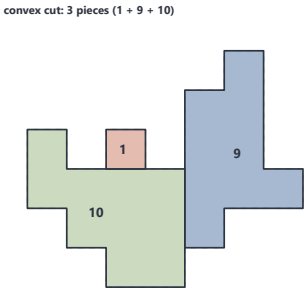


(a) concave vertices (b) minimum cut (c) split and recurse

- Find a concave vertex: a direction reversal with the correct area sign.
- Project v onto each edge; choose the nearest visible edge and insert the intersection point.
- Splice the new node, split the circular list, and recurse on both pieces.

Convex Decomposition Example





- Three orthogonally convex pieces.
- Areas $1, 9, 10$; total area 20.

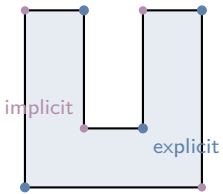
Base Cases and Recursion

Base cases.

- 2 stored points: a rectangle; stop.
- 3 stored points: a monotone / L-shaped piece; stop in the convex variant.
- The explicit variant has no 3-point base case; it continues until no concave vertex remains.

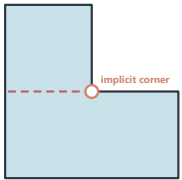
```
cut(v1):
    v2, v3 = v1.next, v1.next.next
    if v3 == v1:
        return [v1, v2]
    if triangle_base and v3.next == v1:
        return [v1, v2, v3]
    c = find_concave(v1)
    if c is None:
        return all_vertices(v1)
    vmin, vertical = find_min_dist(c)
    new = insert_cut(c, vmin, vertical)
```

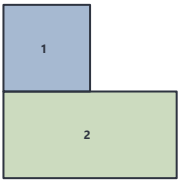
Explicit and Implicit 270° Vertices



- **Explicit** $270°$**:** the reflex corner is a stored even-index vertex.
- **Implicit** $270°$**:** the reflex corner is created by the x-first expansion.
- The same compressed U-shape contains both kinds.
- The convex cut handles the shape; the explicit and implicit cuts handle the corner types.

Implicit Cut of an L-Shape



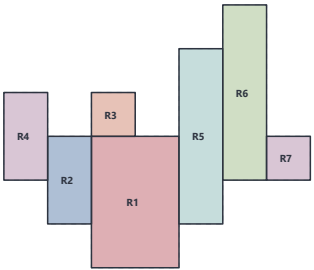


- The implicit corner $(1, 1)$ is reflex.
- A horizontal cut yields rectangles of areas 2 and 1.

Rectangle Output and Canonicality

- A rectangle piece has **2 stored points**: two opposite corners.
- A canonical 2-point list only needs distinct points; a non-degenerate rectangle also needs distinct x and y.
- The three-step algorithm outputs rectangles; the tests assert the two-point form, distinct coordinates, canonicality, and the area.
- For larger intermediate pieces, canonicality and orthogonal convexity are checked on the expanded boundary.

Rectangle Decomposition



- Seven rectangles, areas $1, 1, 2, 2, 4, 4, 6$.
- Total area 20.

Correctness Invariants and Complexity

Invariants.

1. **Area:** the sum of signed piece areas equals the signed area of the input.
2. **Rectangle output:** the three-step algorithm returns two-point rectangles.
3. **Canonicality:** local. Orthogonal convexity is checked on the expanded boundary.

Complexity.

operation	cost
canonicality check	$O(m)$
canonicalization	$O(m)$
concave-vertex scan	$O(n)$
nearest-edge scan	$O(n)$
insert / split	$O(1)$
total decomposition	$O(n^2)$

For $n \leq 100$, quadratic time is preferable to a complicated sub-quadratic

Section 6

Verification

Implementation and Tests

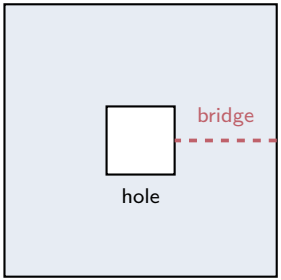
C++20 (xmake)

- `xmake build test_recti`
- `xmake run test_recti`
- **143 test cases**
- **974 assertions**
- **0 failures**
- Canonicalization processes the creation ordering before the canonicality check; raw creation is only an ordering, not a polygon.
- The tests assert the output class: canonicality, orthogonal convexity, and rectangle shape.
- A bridged-hole fixture is cut into three pieces of areas 2, 15, and 10; the area is preserved.

Limitations and Threats to Validity

1. **Canonicality.** It is a local property of the encoding; simplicity is an assumption on the input.
2. **Synthetic inputs.** The evaluation uses generated point sets and hand-built fixtures, not foundry data; the claims are algorithmic, not signoff.
3. **Quadratic scaling.** Decomposition is $O(n^2)$; canonicalization is $O(m)$. For $n \leq 100$ this is appropriate, and sub-quadratic methods are not addressed here.
4. **No optimality claim.** The three-step algorithm returns a rectangle partition, not a minimum-cardinality one.
5. **Robustness.** Integer arithmetic is exact; degenerate input handling beyond the stated invariants is outside the present scope [5].

Polygons with Holes



- A bridge cut from the outer boundary to a hole merges the boundaries into a single cycle; a zero-width bridge gives a weakly simple boundary, a narrow bridge gives a simple one.
- The bridge is traversed in both directions, so the area is unchanged.
- The even-index encoding and the cut algorithms then apply; repeated bridge vertices are handled by the degenerate split.
- This is standard boundary bookkeeping, not specific to the even-index

Section 7

Conclusion

Contributions and Open Problems

Contributions.

- A canonical form with a redundant-vertex criterion.
- An $O(m)$ canonicalization that preserves perimeter and area.
- A three-step rectangle decomposition: convex decomposition, explicit 270° handling, and implicit 270° handling.
- Output-class tests and C++20 verification (143 cases, 974 assertions).

Open problems.

- A formal proof that the compressed predicate characterizes orthogonal convexity.
- Minimum-cardinality rectangle partitioning.
- Cross-language differential testing on random polygons.

Q&A

- Code: `physdes-cpp` under the `luk036` organization.
- Thank you.

References I

[1] N. A. Sherwani, *Algorithms for VLSI physical design automation*, 3rd ed. Boston, MA: Kluwer Academic Publishers, 1999.

[2] A. B. Kahng, J. Lienig, I. L. Markov, and J. Hu, *VLSI physical design: From graph partitioning to timing closure*. Dordrecht: Springer, 2011.

[3] A. B. Kahng, C.-H. Park, X. Xu, and H. Yao, "Layout decomposition for double patterning lithography," in *Proc. IEEE/ACM int. Conf. Computer-aided design (ICCAD)*, 2008, pp. 465–472. doi: 10.1109/ICCAD.2008.4681616.

[4] L. W. Liebmann, "Layout impact of resolution enhancement techniques: Impediment or opportunity?" in *Proc. Int. Symp. Physical design (ISPD)*, 2003, pp. 110–117. doi: 10.1145/640000.640026.

[5] H. Edelsbrunner and E. P. Mücke, "Simulation of simplicity: A technique to cope with degenerate cases in geometric algorithms," *ACM Trans. Graph.*, vol. 9, no. 1, pp. 66–104, 1990.

References II